

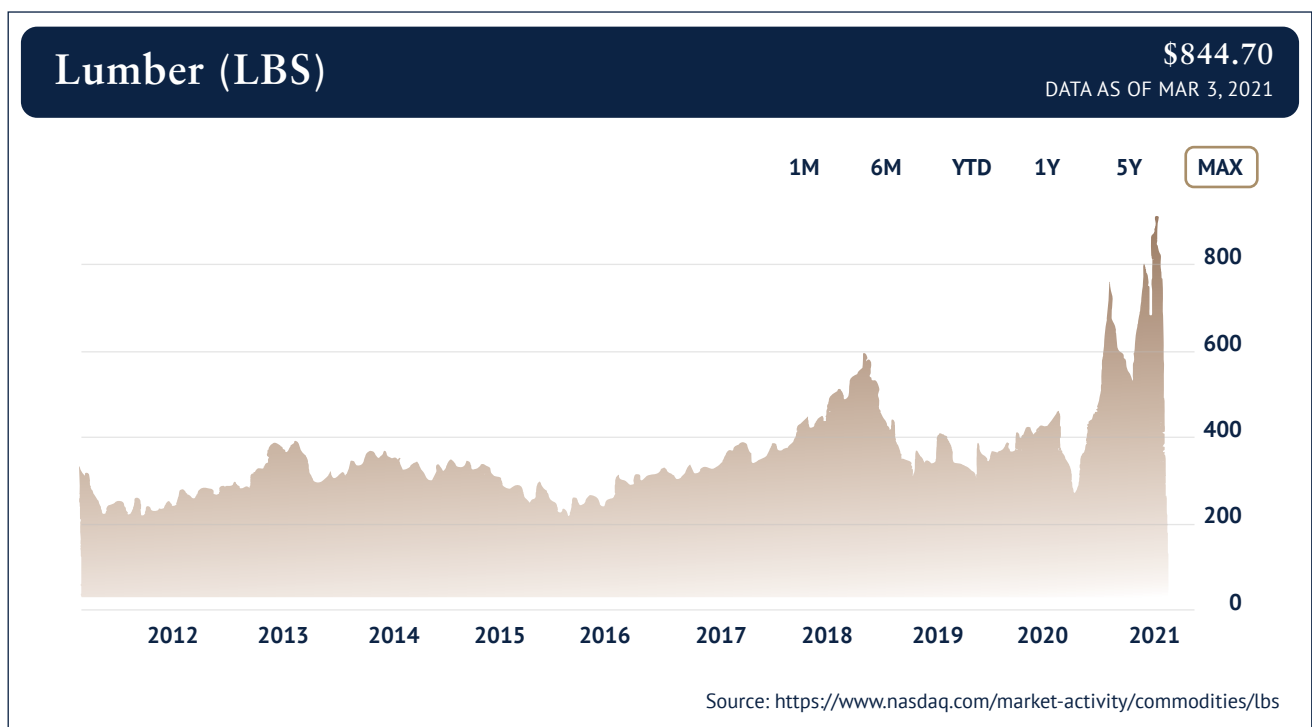


FROM THE RESEARCH ARCHIVES

A Supply-Constrained Commodity Inflation Vector: Lumber



Inflation is beginning to intrude on investors' collective awareness. Until now, the visible mainstream indicators have been absent. Despite financial asset inflation – money, stocks, bonds – no such signs have emerged in the CPI (a highly engineered number). But, in key commodity after key commodity, prices have been increasing markedly. Lumber is one, having more than doubled from pre-pandemic levels a year ago.



Wood Consumption

According to the U.S. Forest Service, domestic wood consumption had been in decline for over two decades. Per capita consumption, which was 83 cubic feet in 1986, declined to 42 cubic feet by 2009. Since then, though, per capita wood consumption increased by almost 25%, to 52 cubic feet in 2017¹, the last year for which there are reliable statistics. This occurred even as paper consumption dropped by over 40% between 1999 and 2017, newsprint capacity declined by 80%, and writing paper consumption dropped by 45%.

Wood is ubiquitous in so many products, that lumber prices find their way into packaging, home prices and more, which will ultimately impact the prices of consumer goods, and is highly inflationary. There is no good substitute for lumber as a construction material or for paper packaging, especially if the goal is to reduce plastic packaging. However, the industry cannot easily respond to increased demand.

¹<https://www.fs.usda.gov/treearch/pubs/58506>

Supply Constraints: Sustainable Forestry Practices

According to the Society of American Foresters, roughly one-half of all U.S. forest land has adopted sustainable cutting practices, and more adopt it every month. It is a massive resource, amounting to 753 million acres. Because of the sustainability provisions of forestry certification programs, forest lands are rigorously monitored by satellite. Ever fewer businesses dealing in lumber will sell wood that is noncertified, meaning that the timberland owner has not adopted sustainable silviculture nor limited the annual cut to the natural growth rate of the forest. The Forest Stewardship Council (FSC) and the Programme for the Endorsement of Forest Certification (PEFC) are imposing new forestry guidelines oriented towards the development of sustainable forest management practices to reduce the annual cut to a rate that can be replaced naturally, i.e., *permanently reducing lumber supply*.

This means that the industry has a reduced capacity to respond to the rising demand for lumber. The proportion of the industry that is certified—and certification can apply both to entities that own timberland and to companies that engage in logging—will almost certainly increase, since buyers are under great pressure to purchase only certified lumber and, in some cases, companies are legally compelled to do so.

Nor is this merely a domestic phenomenon. The U.S. is the fourth largest forest nation, behind Russia, Brazil and Canada, according to Statista, which reports that 31% of total world forest area is now certified. These statistics should be considered in the context of political and social changes that are now in process with respect to environmental sustainability. China and India alone have 2.8 billion people, and a higher standard of living on a sustainable basis that includes eliminating plastics will involve the increased use of paper and wood products.

If forest management is based on the natural replacement rate of the wood, production cannot increase to suit surges in demand, and that changes the character of the business forever. If the policies to replace plastic with paper for packaging continue, it is reasonable to expect sustained high lumber prices and concomitant inflation.

The Trajectory of Process

In the field of energy, non-governmental organizations are combining forces with many institutional investors to compel the reduction of the use of plastic in packaging in favor of more readily biodegradable wood fiber products. This will result in a diminution of plastic production.

In the field of wood fiber, non-governmental organizations are combining forces with large corporate buyers of wood fiber and many institutional investors to compel the reduction in the annual timber harvest to sustainable levels. This has already resulted in a diminution of wood fiber production and will continue to do so.

In the field of monetary policy, the various central banks of the world are promoting maximum stimulus to increase worldwide consumer demand that will strengthen economies and increase employment. This will result in increased demand for packaging materials. Perhaps the packaging might be based on wood fiber or perhaps it will be based on plastics, but in either case, increased economic activity demands increased packaging.

In the field of investing, the well-intentioned thrust towards indexation has never been more powerful. The indexes have minimal exposure to wood fiber companies or energy companies. At the same time, policies are adopted to increase demand for commodities while reducing production of those very same commodities to sustainable levels. One might be tempted to say that such sustainability policies are, quite possibly, not actually sustainable.



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